Solutions: Optimization

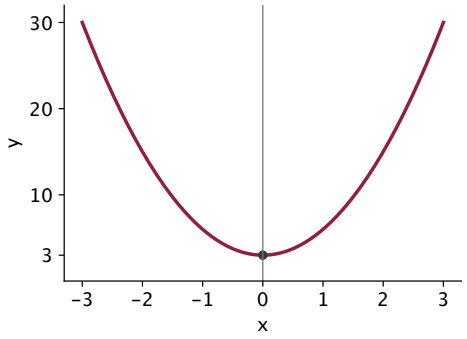
ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 9.2

1. (c) $y = 3x^2 + 3$

$$\frac{dy}{dx} = 6x = 0 \implies x^* = 0$$

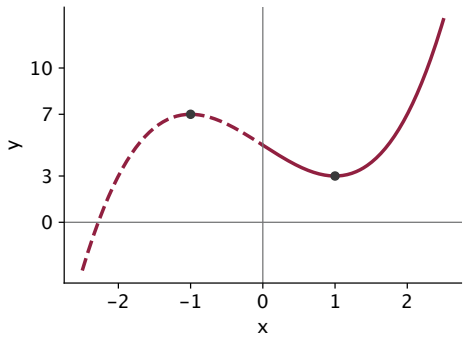
In the immediate neighborhood of 0, $dy/dx < 0$ for $x < 0$ and $dy/dx > 0$ for $x > 0$. The slope changes sign from negative to positive, so the function is decreasing to the left of 0 and increasing to the right: $f(0) = 3$ is a relative minimum. The second derivative confirms this: $d^2y/dx^2 = 6 > 0$.



2. (a) $y = x^3 - 3x + 5$

$$\frac{dy}{dx} = 3x^2 - 3 = 0 \implies x^* = \pm 1$$

The derivative $3(x^2 - 1)$ is negative on the immediate left of 1 (for example at 0.9) and positive on the immediate right (at 1.1), so there is a relative minimum at $x = 1$. It is positive on the immediate left of -1 and negative on the right, so there would be a relative maximum at $x = -1$. But the domain is $[0, \infty)$, so -1 is not admissible. The only stationary value is the relative minimum $f(1) = 3$. The second-derivative test gives the same answer: $d^2y/dx^2 = 6x$, which is $6 > 0$ at $x = 1$.

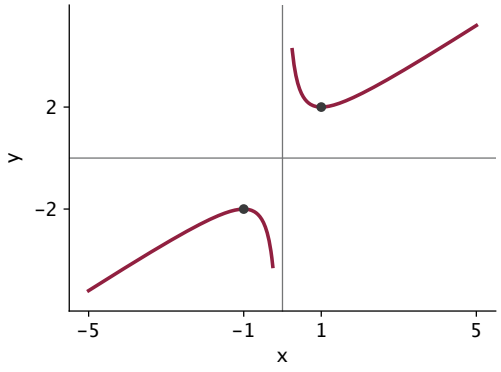


3. $f(x) = x + 1/x$

$$f'(x) = 1 - \frac{1}{x^2} = 0 \;\Rightarrow\; x^* = \pm 1$$

The derivative, $(x^2 - 1)/x^2$, is negative on the immediate left of 1 and positive on the immediate right, so $f(1) = 2$ is a relative minimum. It is positive on the immediate left of -1 and negative on the right, so $f(-1) = -2$ is a relative maximum.

The relative maximum, -2, is lower than the relative minimum, 2. This is not a contradiction: these are *relative* extrema, each compared only with nearby points on its own branch of the graph. The function is not defined at $x = 0$, and the two branches never meet.



4. $T = \phi(x)$

(a) $M = \phi'(x)$ and $A = \phi(x)/x$.

(b) At an extremum of A,

$$A' = \frac{\phi'(x)\,x - \phi(x)}{x^2} = 0 \;\Rightarrow\; \phi'(x^*) = \frac{\phi(x^*)}{x^*}$$

that is, $M = A$.

(c) The marginal curve must cross the average curve at the point where the average curve reaches its extremum (its lowest point for a U-shaped average cost curve).

(d) The elasticity of T is

$$\varepsilon = \frac{\phi'(x)\,x}{\phi(x)} = \frac{M}{A}$$

When $M = A$, the elasticity equals 1.

Exercise 9.4

1. (b) $y = x^3 + 6x^2 + 9$

$$y' = 3x^2 + 12x = 3x(x+4) = 0 \;\Rightarrow\; x^* = 0, -4; \qquad y'' = 6x + 12$$

$y''(0) = 12 > 0$, so $y(0) = 9$ is a relative minimum. $y''(-4) = -12 < 0$, so $y(-4) = 41$ is a relative maximum.

(d) $y = \dfrac{2x}{1-2x}$. By the quotient rule,

$$y' = \frac{2(1-2x) - 2x(-2)}{(1-2x)^2} = \frac{2}{(1-2x)^2}$$

which is never zero. There is no stationary point, so there are no relative extrema. The function is increasing on each side of $x = 1/2$.

2. Let L be the side parallel to the wall and W the two sides perpendicular to it. The netting constraint is $L + 2W = 64$, so $L = 64 - 2W$ and the area is

$$A = LW = (64 - 2W)W = 64W - 2W^2$$

$$\frac{dA}{dW} = 64 - 4W = 0 \;\Rightarrow\; W^* = 16, \quad L^* = 32$$

Since $d^2A/dW^2 = -4 < 0$, this is a maximum. The largest area is $32 \times 16 = 512$ square feet.

3. (a) Yes: $a = 1/3 > 0$, $b = -7 < 0$, $c = 111 > 0$, $d = 50 > 0$, and $b^2 = 49 < 3ac = 111$.

(b) From the demand function, $P = 100 - Q$, so $R = PQ = (100-Q)Q = 100Q - Q^2$.

(c) $\pi = R - C = 100Q - Q^2 - \frac{1}{3}Q^3 + 7Q^2 - 111Q - 50 = -\frac{1}{3}Q^3 + 6Q^2 - 11Q - 50$

(d)

$$\frac{d\pi}{dQ} = -Q^2 + 12Q - 11 = 0 \;\Rightarrow\; (Q-1)(Q-11) = 0 \;\Rightarrow\; Q = 1 \text{ or } 11$$

$\frac{d^2\pi}{dQ^2} = -2Q + 12$ is $10 > 0$ at $Q = 1$ (a minimum) and $-10 < 0$ at $Q = 11$ (a maximum). So $Q^* = 11$.

(e) $\pi(11) = -\frac{1}{3}(1331) + 6(121) - 11(11) - 50 = 334/3 \approx 111.33$.